

Mechanical Engineering CAD Curriculum: Tinkercad to Pre-Fusion

A progressive, checkpoint-based path from zero CAD experience to intermediate parametric modeling. Each stage ends with a project you can finish before moving on. It's designed to cover a wide variety of topics so that you can be prepared for creating more advanced projects later on in the year.

Stage 0: Foundations (before touching software)

- Learn to read a basic engineering drawing: orthographic views, dimensions, units.
- Learn the 3 primary planes (XY, XZ, YZ) and right-hand coordinate system.
- Learn basic units/tolerance concepts: mm vs in, $\pm$ tolerance, fit (clearance vs press).
 - Checkpoint 0: Sketch three views (top/front/side) of a simple object on your desk.

Stage 1: Tinkercad (Solid-based, beginner CAD)

Tinkercad teaches solid modeling through boolean operations (add/subtract shapes) no sketching required, which makes it the right entry point.

Learn

- Placing/moving/scaling primitive shapes
- Grouping (union) and hole objects (subtraction)
- Aligning and snapping
- Workplanes for angled features
- Exporting STL
 - Checkpoint 1a: Nameplate/Keychain: Extruded text on a base plate with a mounting hole.

- Checkpoint 1b: Enclosure Box: A hollow box (union outer shell, subtract inner cavity) with a lid that has a lip/step so it seats correctly. This teaches boolean subtraction precision and wall thickness.
- Checkpoint 1c: Simple Bracket: An L-shaped bracket with 2 mounting holes on each face, correctly dimensioned to hold something (e.g., mount a 9g servo).

Skill gate to do before you move on: you can create a hollow part with consistent wall thickness and place holes precisely by number entry (not eyeballing).

Stage 2: Onshape (free, browser-based, parametric/sketch-driven)

Onshape introduces real parametric CAD: 2D sketches + constraints + features, which is the industry-standard workflow (same logic as SolidWorks/Fusion).

Learn

- 2D sketching: lines, circles, fillets, dimensions
- Sketch constraints (coincident, tangent, symmetric, equal) and fully-constrained sketches
- Extrude, Revolve, Sweep, Loft
- Fillet/Chamfer, Shell
- Mates/assemblies (basic): fastened, revolute, slider
- Feature tree editing (going back and editing a step without redoing the model)
 - Checkpoint 2a: Flat Bracket, redone parametrically: Rebuild your Tinkercad bracket as a sketch (not primitives), fully constrained, then extrude. Change one dimension and confirm the whole part updates correctly.
 - Checkpoint 2b: Revolved Part: Model a pulley, knob, or bottle using Revolve.

- Checkpoint 2c: Two-Part Assembly: A shaft + a bearing/bushing hole that mates with proper clearance fit (not interference). Include a mate that allows rotation.
- Checkpoint 2d: Gear or Cam: Use Onshape's gear/involute tools or a swept profile to make a simple gear pair or cam-follower mechanism. This introduces mechanism motion.

Skill gate to do before you move on: you can fully constrain a sketch (green/solved, no floating geometry) and build multi-feature parts where editing one dimension doesn't break downstream features.

Stage 3: Design-for-purpose concepts (parallel, not software)

Layer these in as you do Stage 2 & 3 projects:

- Tolerancing & fits: clearance vs. press fit, why 3D printers need ~0.20.4mm clearance
- DFM (Design for Manufacturing): draft angles, min wall thickness, overhangs for 3D printing/injection molding
- Basic GD&T vocabulary: flatness, concentricity, datum just enough to read a real drawing
- Material selection basics: why PLA $\neq$ steel $\neq$ aluminum for a given load
 - Checkpoint 3: Take any Stage 2 part and produce a proper dimensioned 2D drawing (orthographic + one dimension set) from it, as if handing it to a machinist.

Stage 4: SolidWorks or Onshape Advanced (parametric CAD, intermediate)

This is the last stage before Fusion full parametric assemblies with real engineering features.

Learn:

- Patterns (linear, circular, mirror)
- Configurations / design tables (same part, multiple sizes)
- Assembly mates beyond basics: cam, gear, path mates
- Sheet metal basics (bends, flat pattern) optional but useful
- Simple FEA/stress check (if available) just to see stress concentrations at fillets vs. sharp corners
 - Checkpoint 4a: Patterned Part: A part with a circular bolt pattern (e.g., a flange) driven by one parameter (bolt count).
 - Checkpoint 4b: Multi-Part Functional Assembly: A small working mechanism e.g., a hinge, a linear slide, or a simple gearbox with 4+ mated parts that move correctly when dragged/simulated.
 - Checkpoint 4c: Capstone: Design something real you actually need (a mount for a cap, or a coaster for a drink). Full flow: sketch to constrained features to assembly to dimensioned drawing.

Suggested pacing

Stage	Time (casual)	Time (focused)
0	1 day	1 day
1 (Tinkercad)	1 week	2-3 days
2 (Onshape core)	3-4 weeks	1-2 weeks

3 (concepts, parallel)	ongoing	ongoing
4 (advanced)	4-6 weeks	2-3 weeks

After this, Fusion 360 is a small jump because you'll already know sketch constraints, features, and assemblies; Fusion mainly adds better simulation, CAM, and generative design on top of what you've learned.